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AI for Bharat Hackathon

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Team Name : Singlebit

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Problem Statement :AI for Media, Content & Digital Experiences
Design an AI-driven solution that helps create, manage, personalize, or distribute digital content more effectively.

Inflomnia: The Digital Brain for Creators
Inflomnia is a **phase-wise AI-powered system** designed to help digital creators reduce operational overload and make better content and monetization decisions.
Most creators struggle due to a **Human Bandwidth Bottleneck**—spending excessive time on emails, moderation, and pricing decisions, leaving limited energy for creativity.
Inflomnia addresses this by introducing **AI-assisted agents** that support creators by automating repetitive tasks and providing context-aware decision insights.

The 4 Pillars of Inflomnia
The Shield (Resilience & Mental Health) — Phase 1 (Core)
Instead of simply showing “views are down,” Inflomnia provides an **Algorithm Weather Report** that distinguishes platform-wide trends from creator-specific performance.
It also assists with spam and toxic comment filtering while surfacing high-value engagement signals.

The Accelerator (Monetization Intelligence) — Phase 1 (Core)
This pillar supports creators in monetization decisions by suggesting **data-backed brand pricing ranges** based on performance and market context.
It also assists with content hooks and structure recommendations for branded content.

The Filter (Safety & Legal Support) — Phase 2 (Expansion)
Acts as an AI-assisted risk and safety layer by scanning brand contracts for potentially unfavorable clauses and identifying suspicious engagement or scam activity on paid posts.

The Digital Twin (Scaling & Automation) — Phase 3 (Roadmap)
The Digital Twin is a long-term capability that learns a creator’s tone and preferences to **assist** with message handling, workflow organization, and content simulations.
Advanced use cases, such as synthetic personas, are planned as future enhancements.

Feature	Old Way	The Inflomnia Way
Workload	You do 90% admin, 10% creating.	AI does 90% admin, you do 100% creating.
Brand Deals	Guessing your price and getting ghosted.	AI negotiates and uses data-backed rates.
Testing	Post and "hope" it works.	**Simulation:** 10 AI agents critique your script *before* you film.

Inflomnia shifts the paradigm from passive creative tools to active **Agentic Infrastructure**.

How is Inflomnia Different?

Active Agents vs. Passive Dashboards

Unlike traditional analytics tools that only display metrics, Inflomnia uses AI agents to take action—such as moderating engagement, assisting pricing decisions, and simulating content outcomes.

Labor Automation (Phase 1 Focus)

Inflomnia digitizes the repetitive 90% of creator business work (moderation, analysis, pricing support), reducing operational overload instead of adding more tools to manage.

Agentic Infrastructure (Phase 2+ Expansion)

As the system evolves, Inflomnia introduces advanced agent-based workflows to support content planning, simulations, and scaled operations.

How Inflomnia Solves the Problem

Breaking the "Human Ceiling"

Inflomnia helps creators scale beyond limited personal time by automating decision support and repetitive operational tasks (initially assisted, not fully autonomous).

Psychological Sustainability

By providing context-aware insights such as Algorithm Weather Reports, Inflomnia reduces stress caused by platform volatility and performance uncertainty.

Professionalizing Creator Businesses (Expansion Phase)

Over time, Inflomnia enables creators and small businesses to adopt structured, data-backed content and monetization practices, improving consistency and confidence.

USPs

Feature	Innovation	Competitive Advantage
10-Persona Simulation	Adversarial AI agents critique scripts.	Predicts audience reaction *before* production spending.
Neural Persona	RAG-based cloning of voice & values.	An autonomous worker that makes real business decisions.
Visual Vibe Audit	Computer Vision for grid analysis.	Matches brands/creators via aesthetic fit, not vanity metrics.

List of features offered by the solution

1. Resilience Engine (The Shield) — Phase 1 (Core)
Helps creators better manage stress caused by platform volatility
Detects whether reach changes are creator-specific or platform-wide
Assists in filtering toxic comments, spam, and bot-driven activity
Provides signals to adjust content workload based on engagement patterns

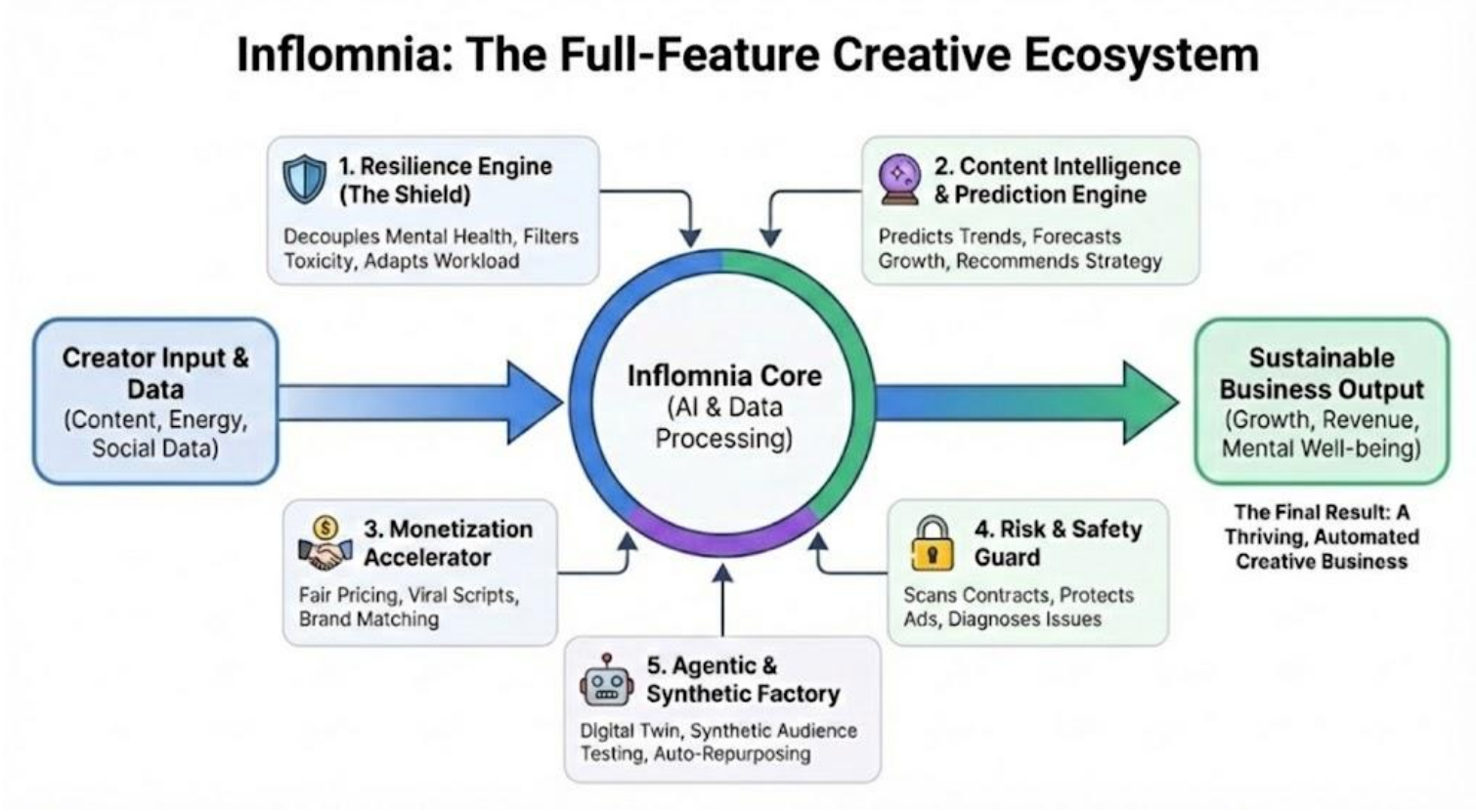
2. Monetization Accelerator (The Accelerator) — Phase 1 (Core)
Helps creators move towards sustainable content monetization
Suggests fair, data-backed pricing ranges for brand deals
Assists with script structure and hook recommendations for branded content
Supports creator–brand matching based on audience relevance and content fit

3. Content Intelligence & Prediction Engine — Phase 2 (Expansion)
Supports creators in deciding what content to create next
Predicts likely-performing formats and topics within a niche
Analyzes emerging trends before they reach saturation
Simulates potential growth trajectories over 3, 6, and 12 months
Recommends data-backed strategic pivots for long-term growth

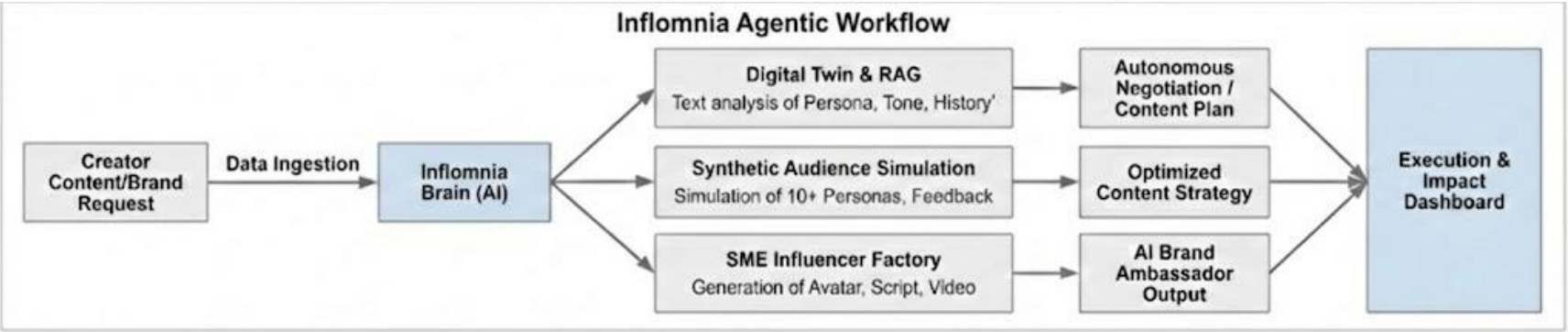
4. Risk & Safety Guard (The Filter) — Phase 2 (Expansion)
Acts as an AI-assisted legal and reputation safety layer
Scans brand contracts to flag potentially unfavorable clauses
Identifies suspicious engagement and scam activity on paid promotions
Helps diagnose technical or policy-related factors affecting reach

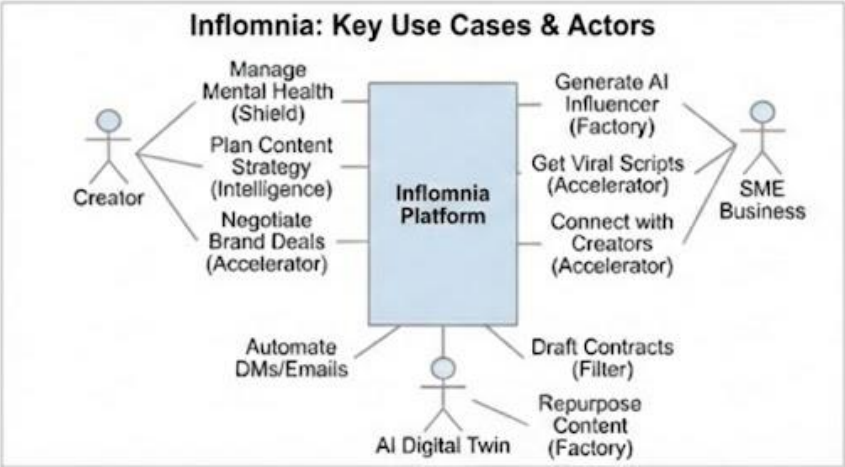
5. Digital Twin & Agentic Systems — Phase 3 (Roadmap)
Learns creator preferences, tone, and workflows over time
Assists with message handling, workflow organization, and content planning

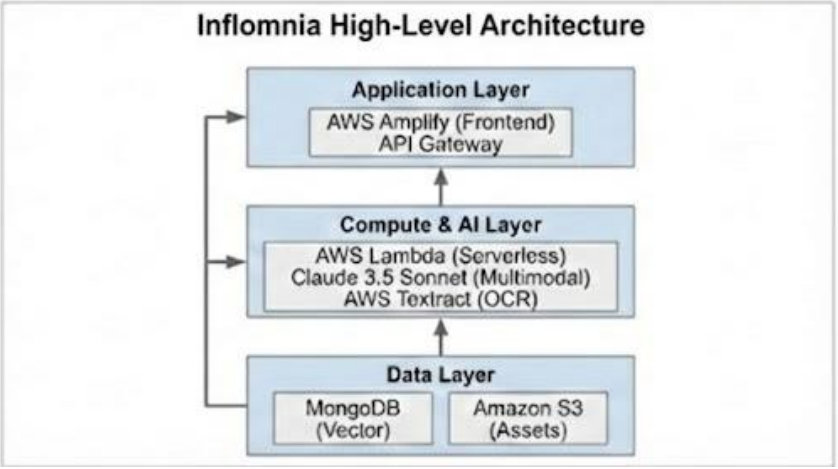
Process flow diagram or Use-case diagram



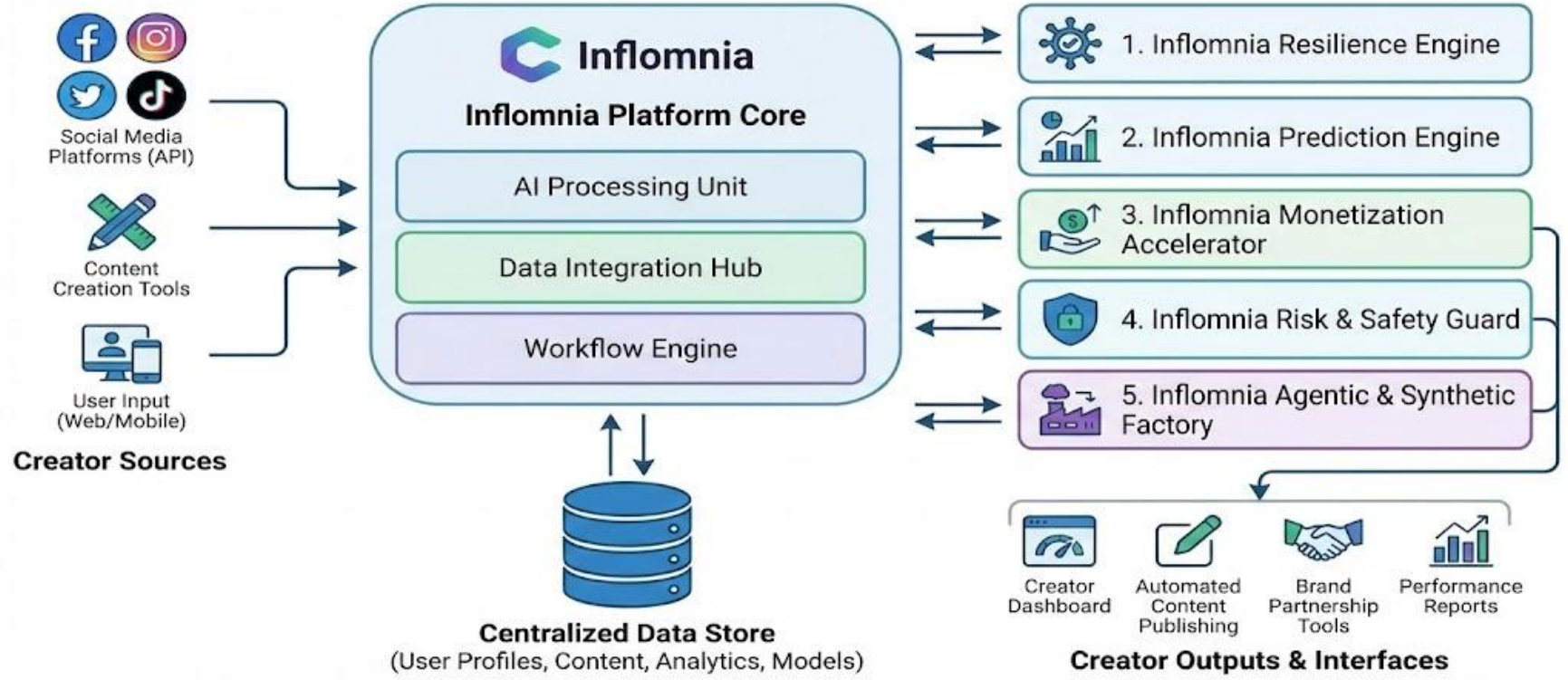
Wireframes/Mock diagrams of the proposed solution (optional)







Architecture diagram of the proposed solution:



Technologies to be used in the solution:

AI & Intelligence

- **Amazon Bedrock (Claude 3.5 Sonnet):** Serves as the primary reasoning engine for complex contract analysis and script simulation.
- **Amazon Bedrock Agents:** Orchestrates multi-step "Agentic Workflows" for automated brand negotiation and task execution.
- **Amazon Bedrock Guardrails:** Ensures "Responsible Design" by implementing PII masking and toxic content filtering.

Compute & Backend

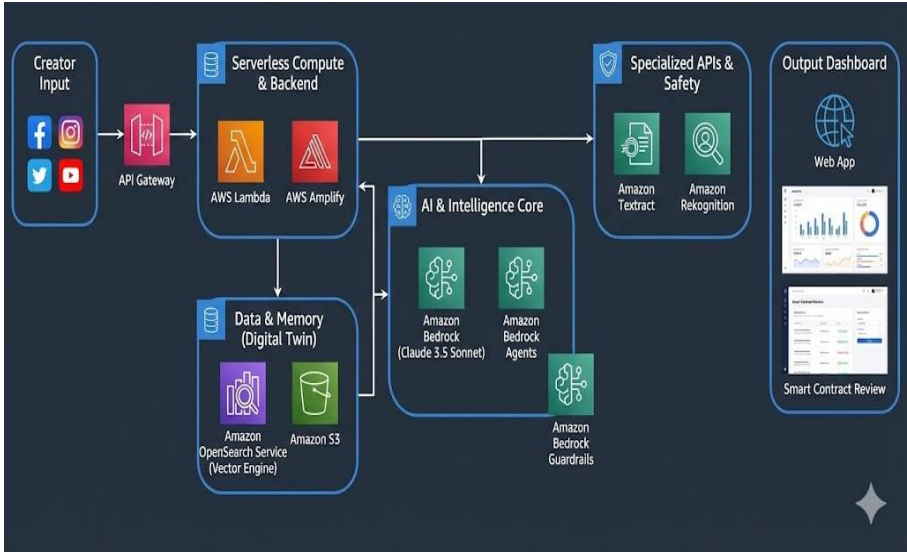
- **AWS Lambda:** Provides serverless, event-driven compute for cost-efficient agent triggers and data processing.
- **AWS Amplify:** Manages the Next.js frontend hosting, user authentication, and CI/CD pipeline.

Data & Memory (The Digital Twin)

- **Amazon OpenSearch Serverless (Vector Engine):** Powering the RAG (Retrieval-Augmented Generation) for the "Digital Twin" long-term memory.
- **Amazon S3:** Scalable object storage for creator media assets, technical blogs, and generated content.

Specialized APIs & Safety

- **Amazon Textract:** Automatically extracts and analyzes clauses from PDF brand contracts for "The Filter".
- **Amazon Rekognition:** Powers the "Visual Vibe Audit" by analyzing media aesthetics and ensuring brand-safe visual content.



Estimated implementation cost:

- **Stage 1 (PoC): $15,000 – $45,000** for core AWS infrastructure and initial agent tuning (2–10 weeks).
- **Stage 2 (Mid-Level): $50,000 – $150,000** for full multi-agent orchestration and large-scale data ingestion (3–6 months).
- **AWS Optimization:** Highlighting the use of **AWS Lambda** (serverless) to ensure the solution has **zero idle costs**, scaling only as the user base grows.

Future Roadmap & Vision

- **Phase 1 (MVP - Current):** Functional integration of **The Shield** and **Accelerator** using **Amazon Bedrock AgentCore** and Claude 3.5 Sonnet.
- **Phase 2 (Expansion - 6 Months):** Full rollout of **The Filter** for legal contract scanning and the **Content Intelligence Engine** for trend forecasting.
- **Phase 3 (Scale - 1 Year):** Launching the **Digital Twin Factory** to handle autonomous brand negotiations and synthetic content production at scale.
- **Vision:** "To turn every aspiring creator in Bharat into a self-sustaining media enterprise by professionalizing the 90% admin work through AI".

Market Opportunity & Viability:

The Addressable Market: India has over **100 million** aspiring creators, yet only a small fraction is professionalized. Your solution targets the "Bharat" segment—creators in Tier 2 and Tier 3 cities who lack access to expensive business managers.

The Monetization Gap: Indian SMEs are desperate for digital presence but cannot afford celebrity influencers. **Inflomnia** bridges this by providing a low-cost "AI Ambassador" through the **Agentic & Synthetic Factory**.

Revenue Model:
Freemium: Free access to **The Shield** (Mental Health/Toxicity filter) to drive user acquisition.

Tiered Subscriptions: Monthly fees for **The Filter** (Legal/Contract scanning) and **Prediction Engine** features.

Enterprise Commissions: Small transaction fees on brand deals successfully audited and negotiated by the AI agents.

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Thank You